IMAGE QUALITY TOOLBOX

SOFTWARE USER MANUAL

Written by	Responsibility-Office-Company	Date	Signature
Romain Degoul	Project team - TELESPAZIO France		
Charlotte Broussou	Project team - TELESPAZIO France		
Verified by			
Sébastien Saunier	Project team - TELESPAZIO France		
Approved			

AMENDMENT RECORD SHEET

ISSUE	DATE	REASON	REDACTEUR AUTHOR
0.1	01/12/2026	Creation of the document	Sébastien S.
0.2	10/03/2026	Draft version of the document for delivery	Romain D.
0.3	16/03/2026	Image update	Romain D.

TABLE OF CONTENTS

TABLE OF FIGURES

1. INTRODUCTION

1.1 Scope

This Software User Manual (**SUM**) describes the Image Quality (**IQ**) toolset developed in the context of the ESA **AI4QC** project. The IQ Toolset is an open-source plugin interfacing with the Quantum Geographic Information System (**QGIS**) software. This plugin is aiming to analyse spatial resolution of remote sensing images. It has been developed for remote sensing Calibration / Validation user as a supporting tool for Quality Control (**QC**)) of remote sensing images.

1.2 Reference Documents

The following is a list of reference documents with a direct bearing on the content of this proposal. Where referenced in the text, these are identified as [RD-n], where 'n' is the number in the list below:

Reference	Title
[AD 1]	"CAL/VAL Geometry framework and AI4QC challenge", TP UK/AG/25/02962
[RD 1]	QGIS Software – PyQGIS Developer Cookbook: https://docs.qgis.org/3.34/en/docs/pyqgis_developer_cookbook/
[RD 2]	QGIS Software – Processing Toolbox Manual: https://docs.qgis.org/3.34/fr/docs/user_manual/processing/
[RD 3]	Image Quality Toolset – GitLab Repository: https://gitlab2.telespazio.fr/isl/tim/edap/image-quality-tool
[RD 4]	QGIS Software – Processing Toolbox Manual: https://docs.qgis.org/3.34/fr/docs/user_manual/processing/toolbox.html

1.3 Acronym list

AI4QC .... AI for Quality Control
ESF .... Edge Spread Function
IQ .... Image Quality
MTF .... Modulation Transfer Function
QC .... Quality Control
QGIS .... Quantum Geographic Information System
ROI .... Region Of Interest
SNR .... Signal to Noise Ratio
SUM .... Software User Manual

2. DESCRIPTION

2.1 SCOPE OF THE TOOL

A fundamental function of a remote sensing sensor relies on its ability to discern ground objects in agreement with mission design / mission specifications. This notion is addressed with the measurement of spatial resolution. From image user perspective, it requires knowledge of image spatial resolution metrics strongly related to image specification parameters.
The IQ Toolset provides a QGIS plugin and a command-line interface to perform image quality assessments from Regions of Interest (**ROIs**) defined on an input image.
The ROIs image is process to derive Modulation Transfer Function (MTF), Signal to Noise Ratio (**SNR**) related image quality measures. Results can be export / saved into a standardized report.
There are three methods on board the QGIS IQ Toolset, these are the following ones:

- **MTF Knife Edge Method**
- **MTF Pulse Method**
- **SNR Method**

The selected image ROI is method specific. Both usage modes share the same underlying algorithms:

Mode	Entry point
QGIS Plugin	Processing Toolbox → ImageQualityToolset
Command Line	image_quality_toolset/scripts/mtf_computer.py

2.2 PROCESSING ALGORITHM OVERVIEW

2.2.1 MTF Knife Edge Method

The MTF Knife Edge method estimates the Modulation Transfer Function of an image displaying a sharp image transition between two uniform areas (as close as possible to a step edge function). The associated processing workflow is shown in the figure just here below.

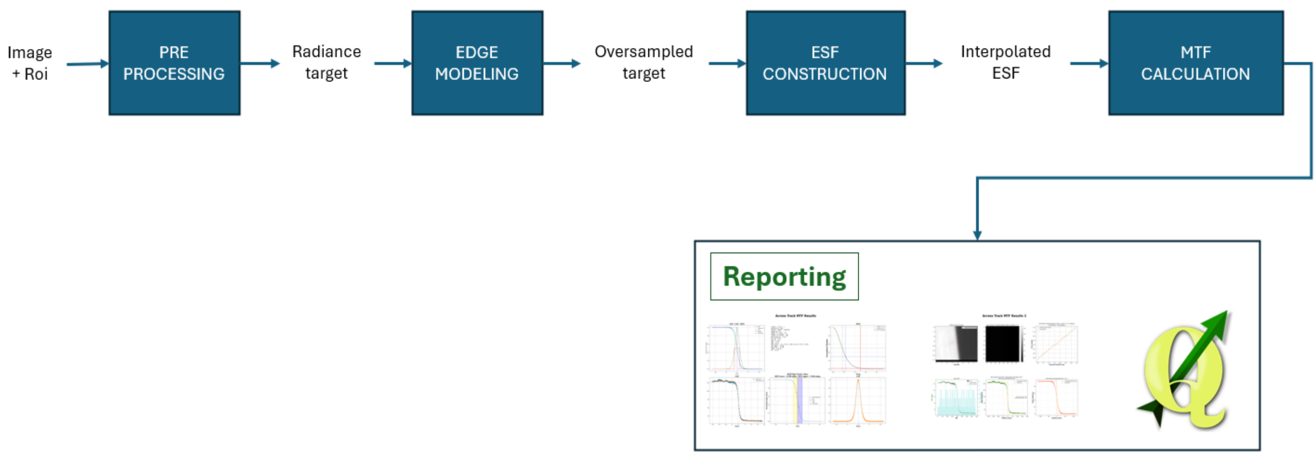


Figure 1 – IQ Toolset – MTF Knife Edge workflow.

A quick description of function involved in the processing is listed in the following table.

Table 1: IQ Toolset – MTF Knife Edge functions.

Functions	Description
Pre Processing	• Convertion to radiance • Configuration Checking

Edge Modelling	• Determine the location of the edge and its angle with respect to the Along-track (AL) and Across-track (AC) directions.
Edge spread function (ESF) construction	• Compute oversample ESF (non-equally spaced) • Apply selected ESF model
Calculation of MTF	• Compute LSF (numerical differentiation) • Normalize LSF • Fast Fourier Transform on the normalized LSF
Reporting	• Spatial Resolution Analysis report (png file x2) • GeoPackage (.gppkg)

An image and a selected ROI are provided as inputs to the algorithm. The ROI is first converted in radiance, oversampled by a chosen factor and then rotated according to the input edge orientation angle.

Inflexions points are retrieved by applied a linear regression line by line or column by column according to the edge orientation. Edge inflection points are estimated by applying a linear regression line-by-line or column-by-column, depending on the edge orientation. The non-equally spaced Edge Spread Function (ESF) is then obtained through statistical binning operations (e.g., mean values within bins). An ESF model is subsequently fitted to the raw ESF, using either parametric or non-parametric approaches.

Five ESF models are available for parametric fitting:

Table 2 - Proposed MTF Models - Parametric method.

Model	Key	Description
Sigmoid	sigmoid	Standard sigmoid function
Hyperbolic tangent	esf_tanh	tanh-based model
Fermi function	esf_fermi	Fermi-Dirac distribution
Gaussian exponential	esf_gauss_exp_param	Parametric Gaussian-exponential model
Erf	Esf_erf	Gaussian Error Function

And the 2 following ESF models are available for non-parametric fitting:

Table 3 - Proposed MTF Models - Non Parametric method.

Model	Key	Description
Loess	esf_loess	Local polynomial regression combining local regression models
Polynomial	esf_polynomial	Polynomial regression model

Several quality metrics are derived from the ESF, including the Signal-to-Noise Ratio (SNR) of the method, relative edge response (RER), Half Edge Extent (HEE),and the R2 error. The interpolated ESF is then differentiated to obtain the Line Spread Function (LSF). From the LSF, the Full Width at Half Maximum (FWHM) is computed. Finally, a Fast Fourier Transform (FFT) is applied to the LSF to obtain the Modulation Transfer Function (MTF). From the resulting MTF curve, several metrics are extracted, including the MTF at Nyquist frequency, MTF at 30% frequency, and MTF at 50% frequency.

2.2.2 MTF Pulse Method

```
TO BE COMPLETED
Same as knife edge method, except model is different
```

2.2.3 SNR Method

The "SNR Method" processing is used to compute SNR @ the mean image radiance. SNR Histogram method, mostly used in the community is complemented with SNR Variogram method. SNR variogram algorithm method set up is more complex, in return it provides more consistent SNR results.

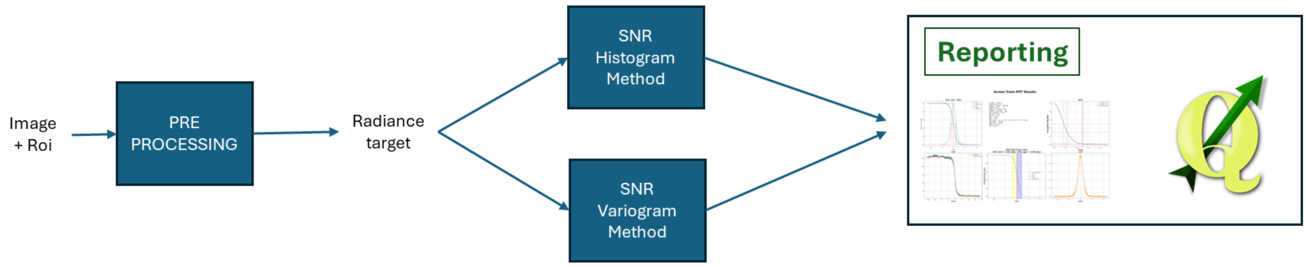


Figure 2 – IQ Toolset – SNR workflow.

Pre processing involves the conversion from digital number to radiance and also configuration checking.

Histogram Method & Variogram Method discussed in ATBD document are proposed.

As reporting, beside GeoPackage file, a panel (figure) is generated and it displays the following items:

- Selected ROI Image
- Image Histogram plot
- SNR Histogram plot
- Variogram plot

3. INSTALLATION PROCEDURE

3.1 INTRODUCTION

The complete installation procedure is detailed in the link bellow: https://gitlab2.telespazio.fr/isl/tim/edap/image-quality-tool/-/blob/main/README.md?ref_type=heads

3.2 PREREQUISITE

The requirements are:

- QGIS (version 3.40 Bratislava): Download · QGIS Web Site
- Python environment compatible with QGIS plugin system

3.3 AUTOMATIC INSTALLATION

```
[Not Working - Planned for the end of the project]
```

QGIS supports integration of plugins written in Python.

Steps:

1. In QGIS, go to Plugins → Manage and Install Plugins…
2. Search for "image quality toolset".
3. Run the installation.

3.4 MANUAL INSTALLATION

First, you need to retrieve the code.

If you have Git access, you can clone the repository with:

git clone https://gitlab2.telespazio.fr/isl/tim/edap/image-quality-tool

Otherwise, you can retrieve the main branch (stable) and develop (unstable) from zip files. It is strongly suggested to start with the main branch since develop is under development.

Then, copy the *image_quality_toolset* directory into your machine's QGIS user plugins directory.

>> Warning:
image_quality_toolset directory is inside *image-quality-tool* directory

```
On Windows:
C:\Users\<username>\AppData\Roaming\QGIS\QGIS3\profiles\default\py
thon\plugins
```

(Replace <username> with your username, and default with your profile name if it's different.)

```
On Linux:
$HOME/.local/share/QGIS/QGIS3/profiles/default/python/plugins/
```

(Replace default with your profile name if it's different.)

To find out your profile name in QGIS, go to Preferences > User Profiles to see which one is selected.

Then, in Plugins > Manage and Install Plugins > Installed, the plugin “***ImageQualityToolset***” should appear. Check the box to enable it.

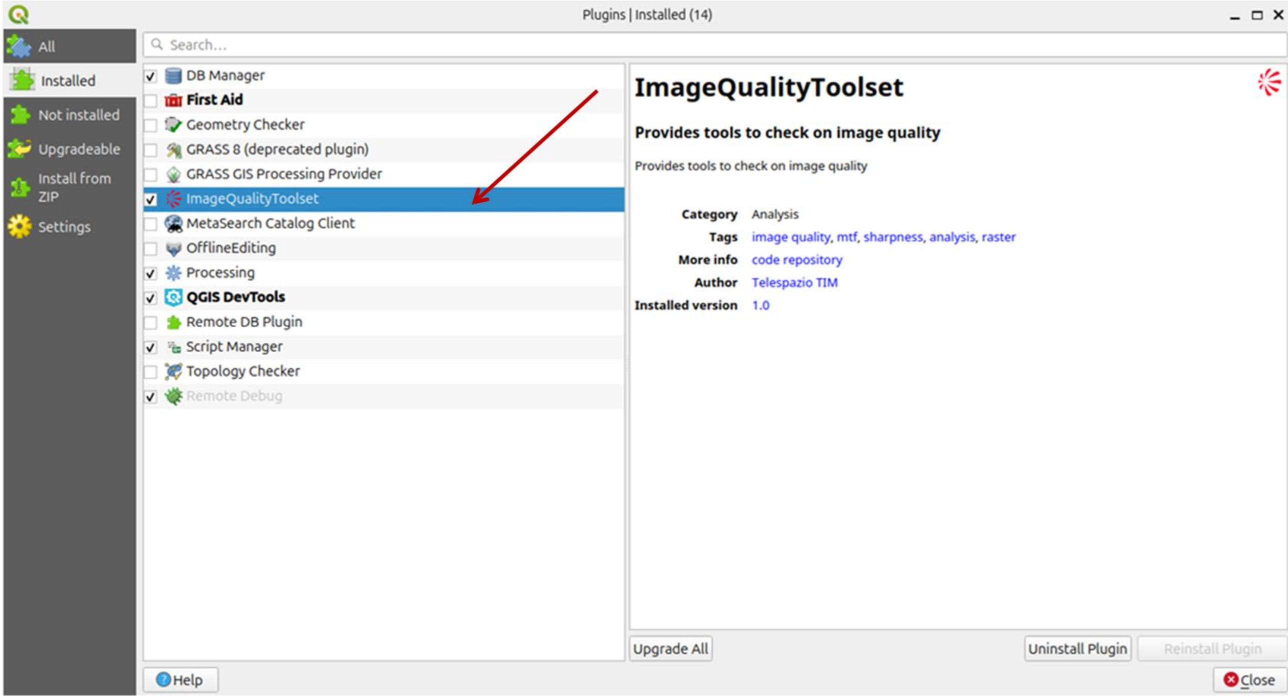


Figure 3 – Find ImageQualityToolset plug-in in the extensions panel.

After extension installation, the plugin should now be available. Also, to access this panel select Processing > Toolbox

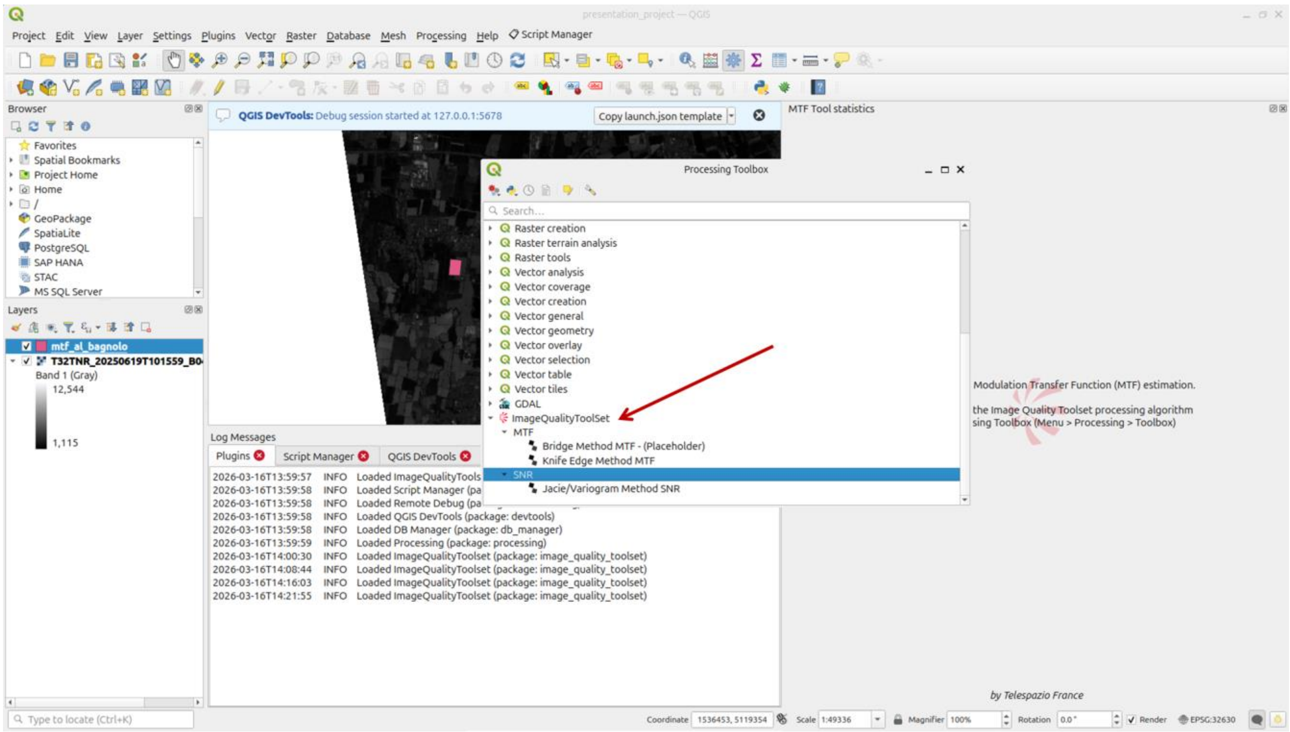


Figure 4 – Processing Toolbox and IQ Tool.

As shown in previous figure, in the processing list select ImageQualityToolset, all available methods appear below. Methods available are:

- MTF Knife Edge Method,
- MTF Pulse Method,
- SNR Method.

4. PLUGIN USAGE

4.1 AVAILABLE METHODS

In the Processing Toolbox, 3 methods are available:

- MTF Knife Edge Method
- MTF Bridge Method
- SNR Method

All these methods need an image and a ROI to work.

4.2 IMAGE SELECTION

To select an image you can drop an image into the QGIS interface or use Layer > Add Layer > Add Raster Layer… and then select an image as Raster dataset.

4.3 ROI PREPARATION

4.3.1 ROI Creation

To create an ROI use Create Layer > New Shapefile Layer.

Then a window appears in which the following fields must be completed:

- File name: name of the ROI, a path can also be selected using ...
- Geometry type: pick Polygon

Other fields are not mandatory.

4.3.2 ROI Edition

Use Toggle Editing then Add Polygon Feature.

Then select the 4 points on the image to make the expected ROI.

Use right-click to validate and press OK (id is optional).

4.4 RUNNING THE METHOD

In the processing toolbox, by double click, select a method in the list, list includes the three methods:

- MTF Knife Edge Method
- MTF Pulse Method
- SNR Method

A window shall appear, and following parameters must be filled for MTF Knife and Edge Method and MTF Pulse Method:

- Radiometric Scale,
- Radiometric Offset,
- Sampling Factor,
- Orientation Angle,
- Pixel Margin,
- MTF Profile,
- ESF Model,
- Raster Layer,
- Band Number,
- Aera of interest.

Static / Dynamic parameters must also be completed depending on the method.

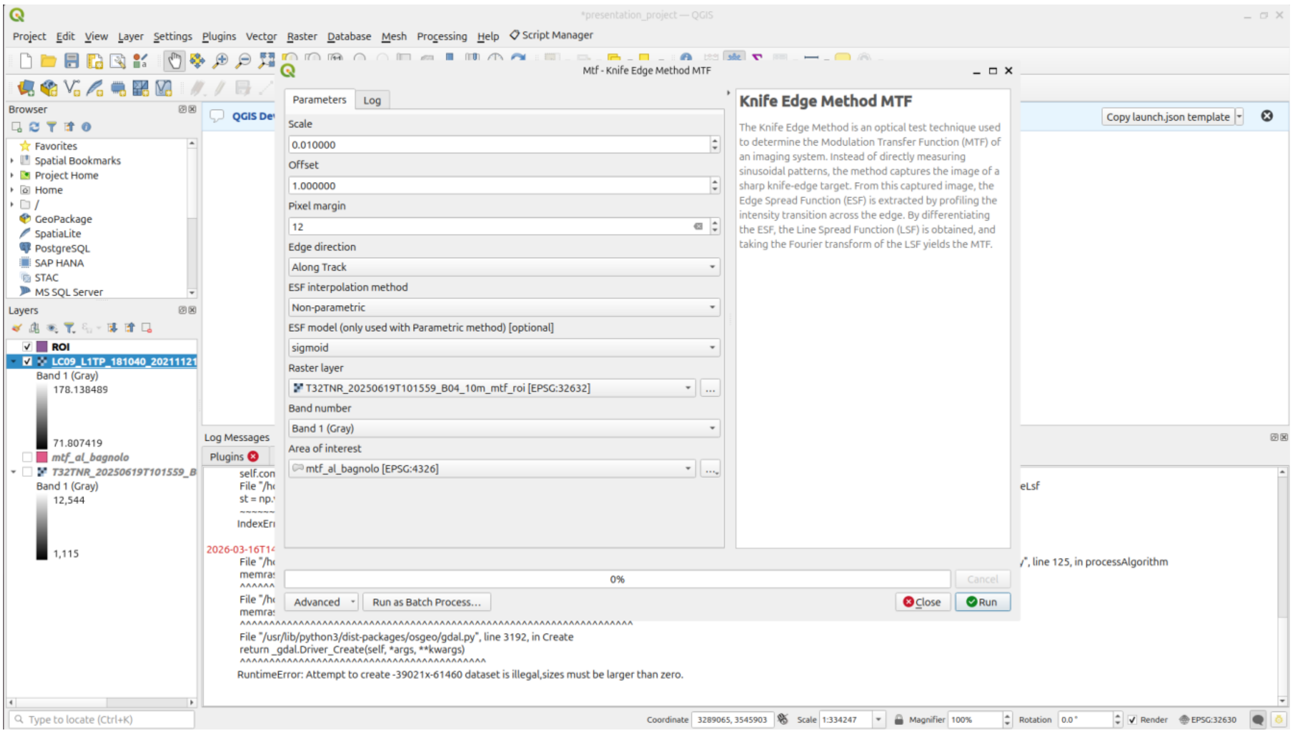


Figure 5 – MTF Knife Edge Estimator, input panel.

Figure 6 – MTF Bridge Estimator, input panel (to do)

And for SNR Method:

- Window Size,
- SNR Precision,
- L min (radiance minimum),
- L max (radiance maximum),
- GSD (Ground Sample Distance in meters),
- Raster Layer,
- Band Number,
- Area of Interest.

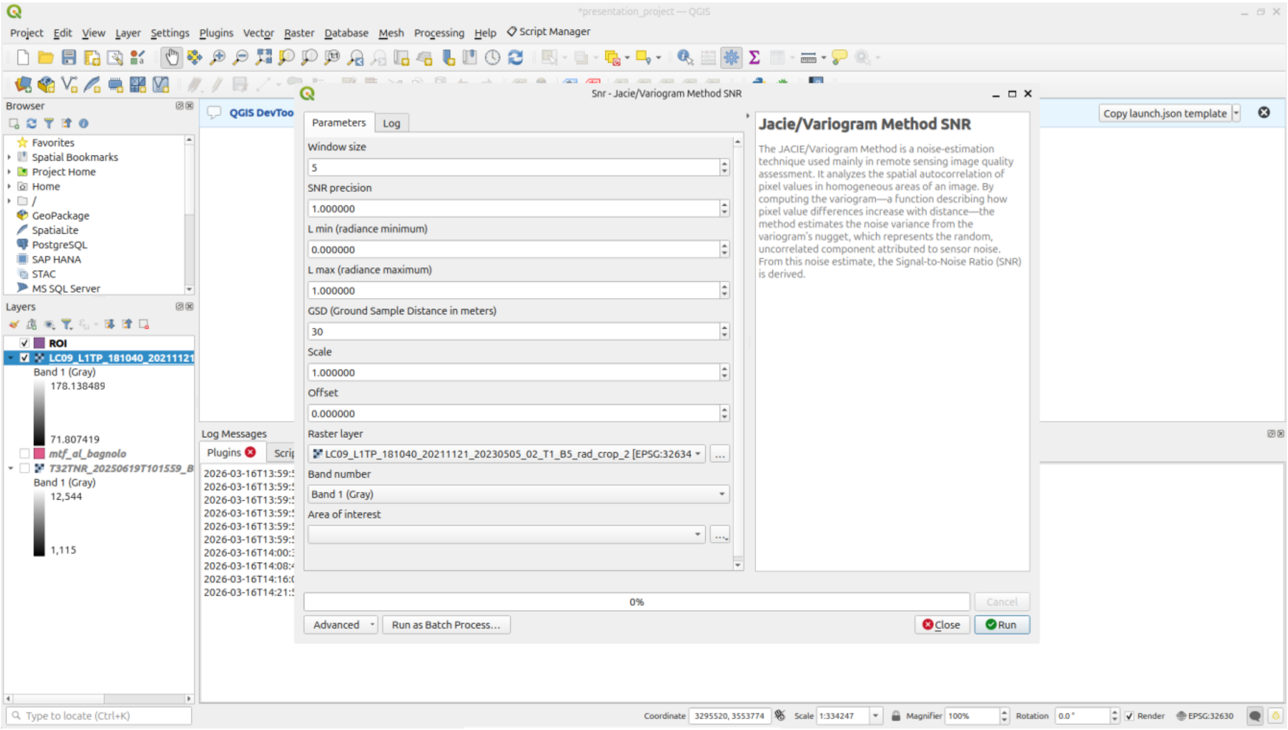


Figure 7 – SNR Estimator, input panel.

```
[Future Update: Export Function Dedicated field in the form (MTF / SNR),
in case directory path indicated; report is saved. ]
```

5. OUTPUTS

5.1 MTF Processing Report (Edge / Bridge)

The MTF Estimator produces two levels of output, depending on the user's needs. A synthetic view is returned directly in QGIS for immediate result inspection, while a detailed export option provides an in-depth breakdown of each processing step for thorough validation and reporting.

5.1.1 Synthetic view

When the MTF Estimator finishes processing, a synthetic view of the results is displayed directly within the QGIS software interface. This view provides an immediate, at-a-glance summary of the main outputs without requiring any export step.

The synthetic view includes:

ROI Extract — A visual display of the selected Region of Interest, allowing the user to verify that the input area was correctly defined and that the edge is properly captured.

Detected Points — The subpixel edge points detected by the algorithm, overlaid on the ROI, enabling quick visual validation of the edge detection accuracy.

Synthetic Information — A condensed summary of the key results from the ESF, LSF, and MTF computations, including the main image quality metrics (MTF@Nyquist, MTF50, MTF30, SNR, RER, FWHM, etc.).

This in-QGIS view is intended as a rapid quality check, allowing the user to assess the validity of the processing and identify any obvious issue before proceeding further.

For a more detailed analysis of each processing step, refer to Part 4.1.2, which describes the export mode and its extended set of diagnostic plots.

5.1.2 Outputs export option

The output export option consists of two panels, each containing six plots. These plots provide detailed information on the target, the Edge Spread Function (ESF), the Line Spread Function (LSF), the Modulation Transfer Function (MTF), and the computation of key image quality metrics.

They allow the user to summarize the processing results and to validate the reliability of the method.

5.1.2.1 First panel

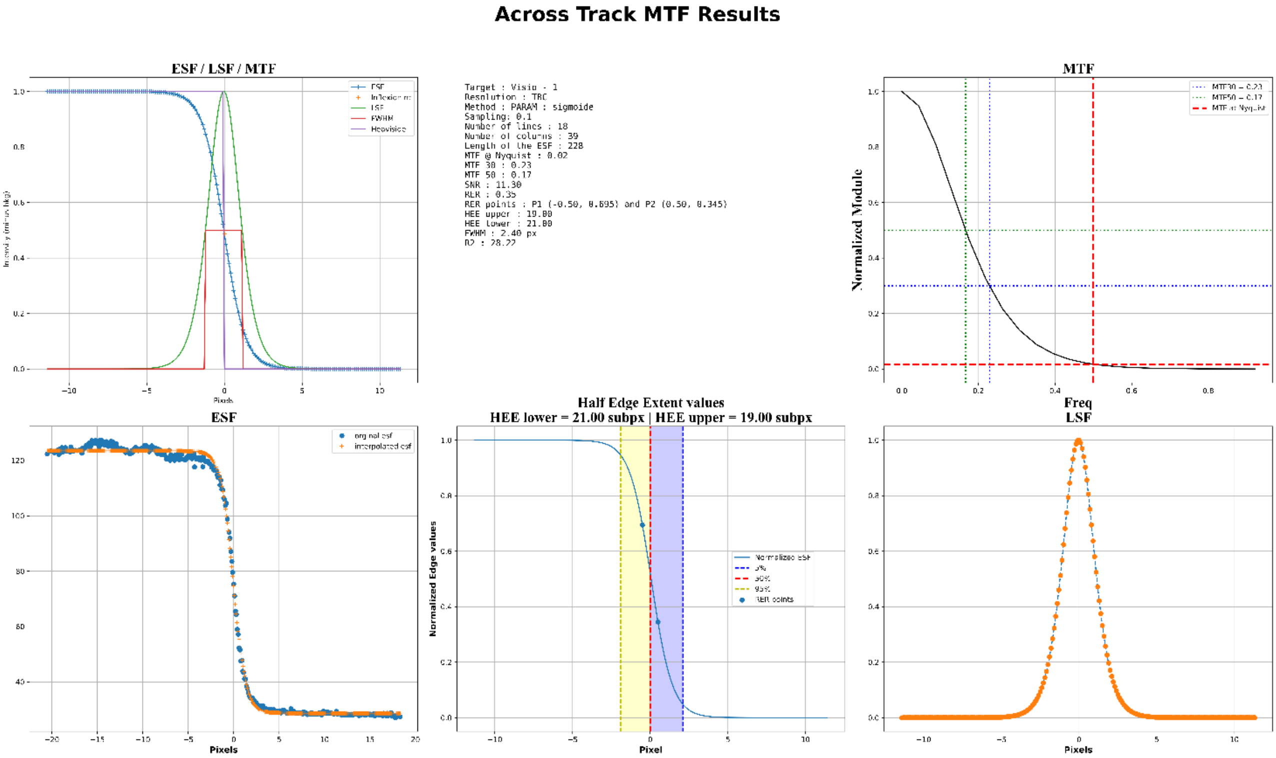


Figure 8 – MTF Knife Edge Method Panel 1.

<u>Plot 1: Global Processing Overview</u>

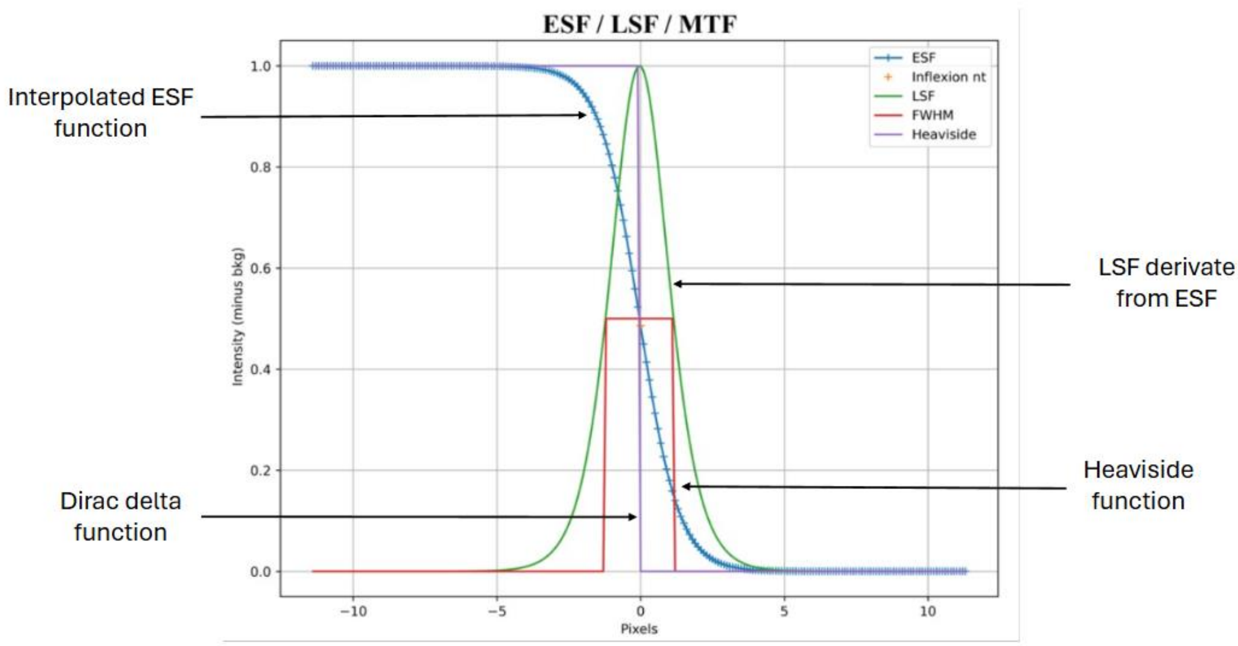


Figure 9 - Global Processing Overview.

This plot provides a global overview of the main processing steps, including ESF, LSF, and MTF computations.

It allows rapid identification of potential anomalies or processing errors.

All functions are normalized and centered around the inflection point of the ESF.

Plot 2: Summary of Input Parameters and Metrics

```
Target : Visio - 1
Resolution : TBC
Method : PARAM : sigmoide
Sampling: 0.1
Number of lines : 18
Number of columns : 39
Length of the ESF : 228
MTF @ Nyquist : 0.02
MTF 30 : 0.23
MTF 50 : 0.17
SNR : 11.30
RER : 0.35
RER points : P1 (-0.50, 0.695) and P2 (0.50, 0.345)
HEE upper : 19.00
HEE lower : 21.00
FWHM : 2.40 px
R2 : 28.22
```

Figure 10 - Summary of Input Parameters and Metrics.

This plot summarizes the main image characteristics, input parameters, and computed performance metrics, including:

- Target: input image data.
- Resolution: spatial resolution of the input data. This information is not retrieved by the algorithm and must be provided by the TBC.
- Sampling: oversampling factor.
- Number of lines: number of rows in the selected ROI.
- Number of columns: number of columns in the selected ROI.
- Length of the ESF: number of samples used to compute the LSF and MTF.
- MTF @ Nyquist: MTF value at the Nyquist frequency.
- MTF 50: spatial frequency at 50% of the MTF.
- MTF 30: spatial frequency at 30% of the MTF.
- SNR: Signal to Noise Ratio value.
- RER: Relative Edge Response value.
- RER points: points used to compute RER.
- HEE upper / HEE lower: Half Edge Extent values above and below the ESF inflection point, respectively.
- FWHM: Full Width at Half Maximum value of the LSF.
- R2: normalized L2 fitting error obtained after ESF modelling.

Plot 3: MTF Validation

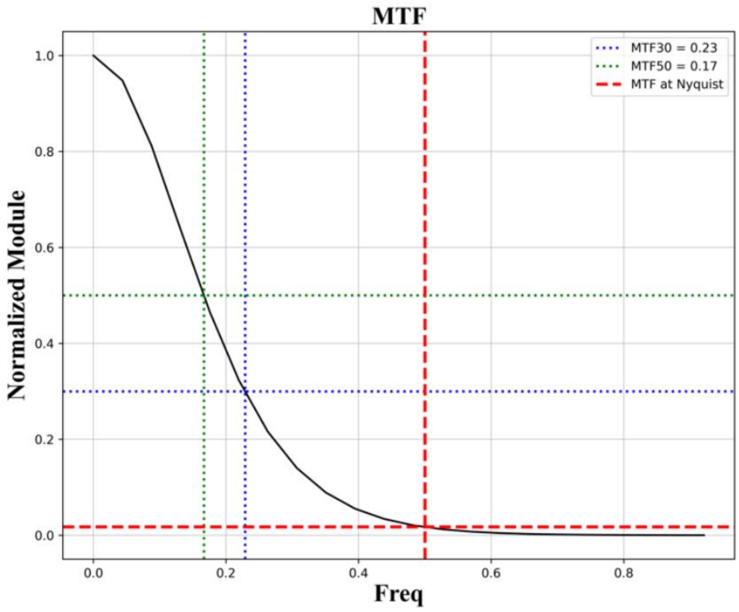


Figure 11 - MTF Validation.

This plot illustrates the MTF curve and highlights the points used to compute:

- MTF at 30%,
- MTF at 50%,
- MTF at the Nyquist frequency.

It enables visual verification of the MTF estimation procedure.

Plot 4: ESF Interpolation

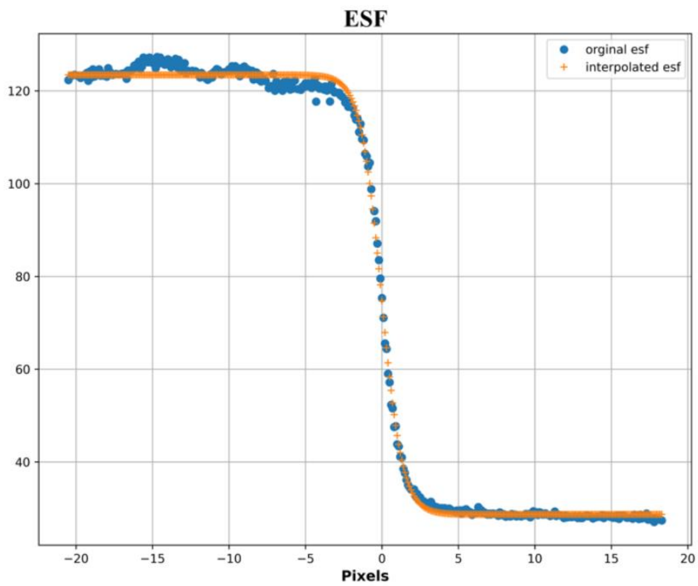


Figure 12 - ESF Interpolation.

This plot displays the ESF samples before and after interpolation.
It illustrates the effect of the parametric or non-parametric processing applied to the raw data.

Plot 5: HEE and RER Determination

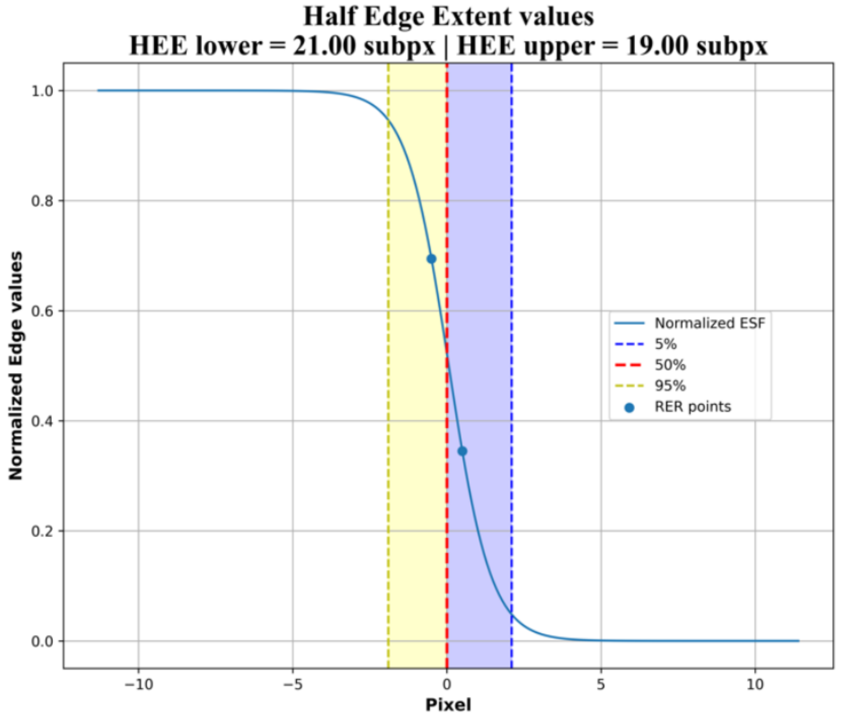


Figure 13 - HEE and RER Determination.

This plot shows the regions used to compute the Half Edge Extent (HEE) and the points involved in the calculation of the Relative Edge Response (RER) on the ESF.
It allows verification of the consistency of these measurements.

Plot 6: Line Spread Function

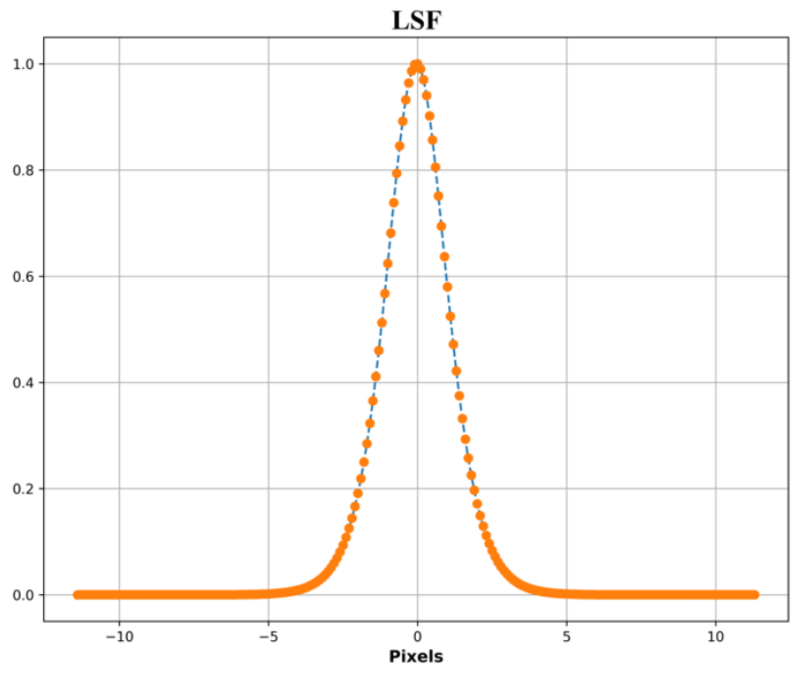


Figure 14 - Line Spread Function.

This plot presents the normalized and centered Line Spread Function (LSF).

5.1.2.2 <u>Second Panel:</u>

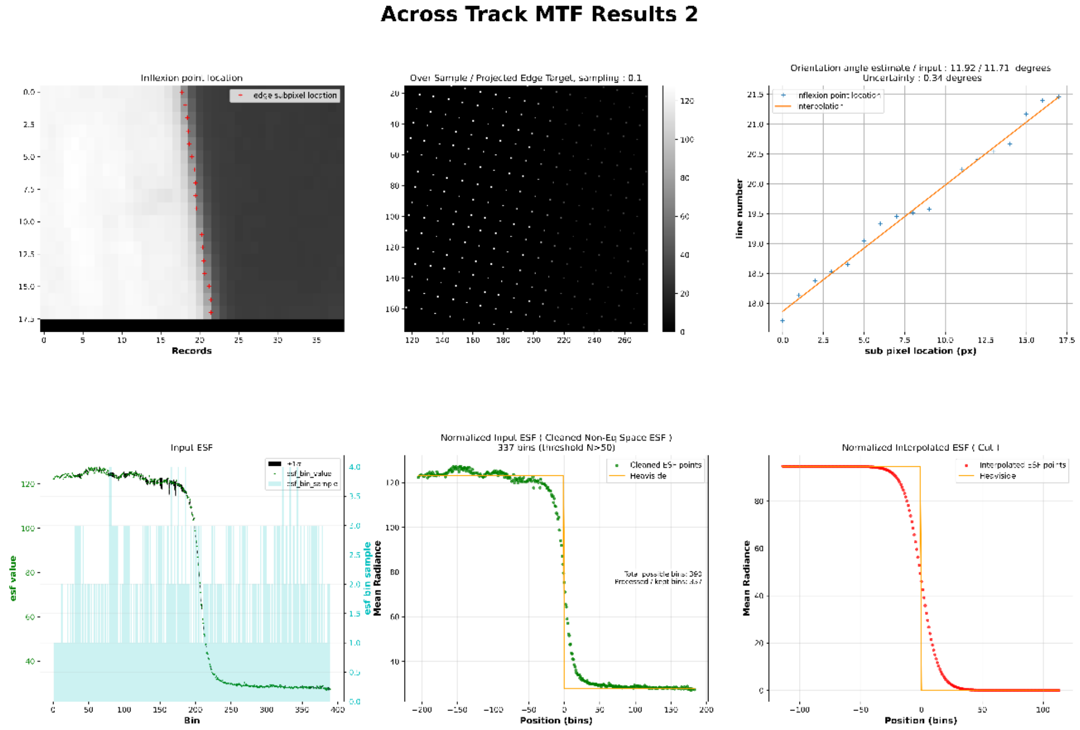


Figure 15 - MTF Knife Edge Method Panel 2.

<u>Plot 1: ROI and Edge Detection</u>

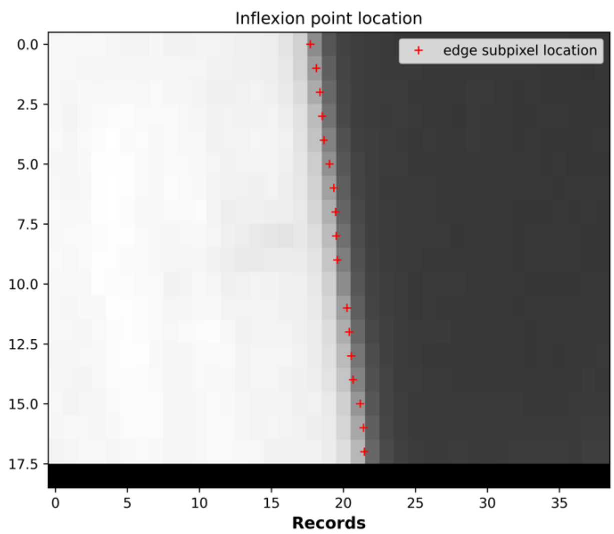


Figure 16 - ROI and Edge Detection.

This plot displays the selected Region of Interest (ROI) and the subpixel edge location detected by the algorithm.

It allows verification of:

- The validity of the ROI selection,
- The accuracy of edge detection.

Plot 2: Oversampled and Rotated Point Cloud

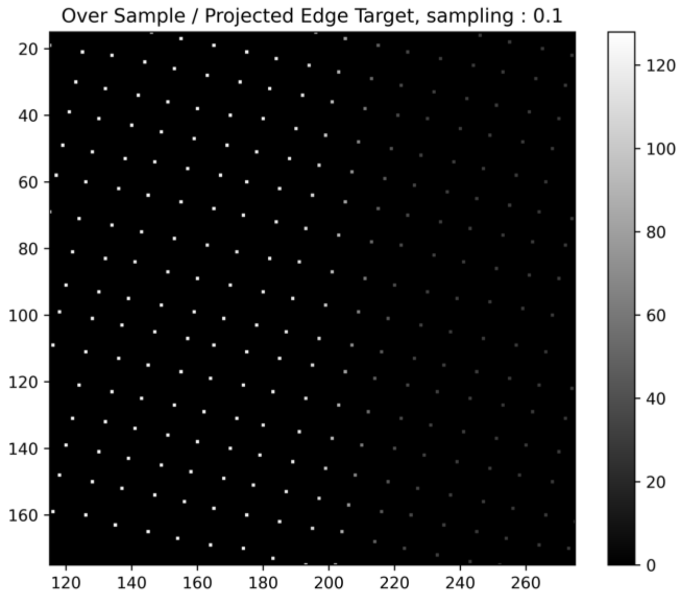


Figure 17 - Oversampled and Rotated Point Cloud.

This plot shows the cloud of samples obtained after oversampling.
The associated diagram represents the intensity value of each sample, where:

- low values correspond to the dark side,
- high values correspond to the bright side.

This visualization illustrates the radiometric transition across the edge.

Plot 3: Edge Orientation and Interpolation

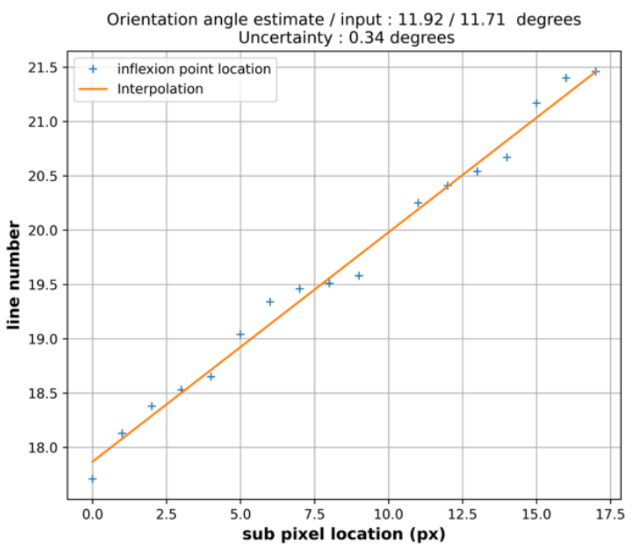


Figure 18 - Edge Orientation and Interpolation.

This plot presents the linear interpolation of edge samples.
The plot title indicates:

- the estimated edge orientation angle,
- the user-defined input angle.
- The uncertainty on the calculated angle.

This comparison allows correction of the input angle if necessary.

Plot 4: ESF Statistical Characterization

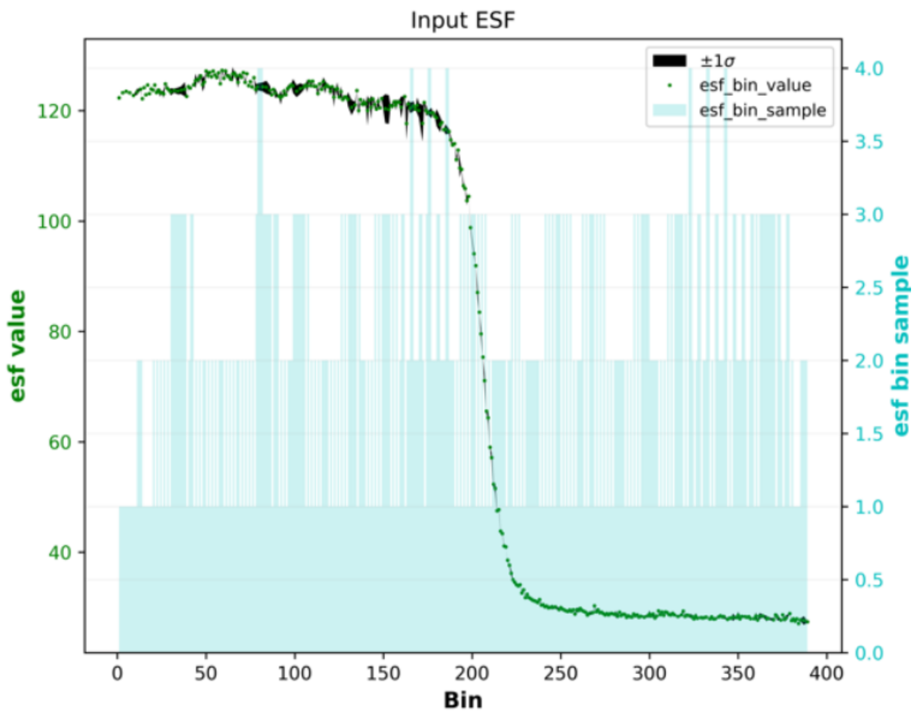


Figure 19 - ESF Statistical Characterization.

This plot displays the ESF value, ESF bin sample and ESF std.
Specifically:

- ESF value: mean value of each ROI row, used to reconstruct the ESF.
- ESF bin sample: number of samples contributing to each ESF value.
- ESF std: standard deviation of the samples used to compute each ESF value.

These indicators quantify the statistical reliability of the ESF estimation.

Plot 5: Normalized input ESF

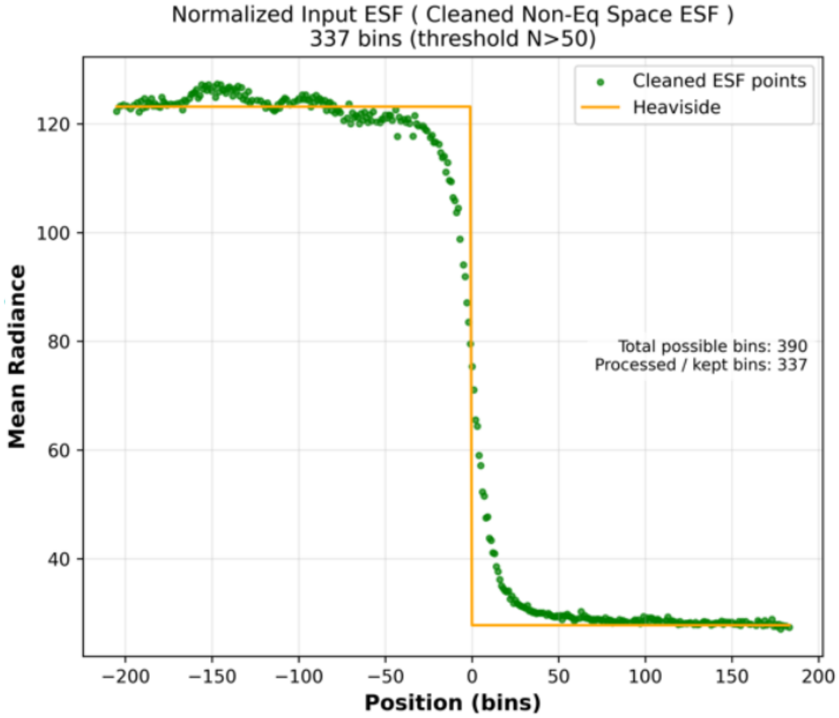


Figure 20 - Cleaned non equally spaced ESF.

This plot displays the normalized and centered ESF clean cloud used to compute the interpolated ESF. The total number of bins and the number of bins used before the ESF interpolation are also shown.

<u>Plot 6: Interpolated ESF</u>

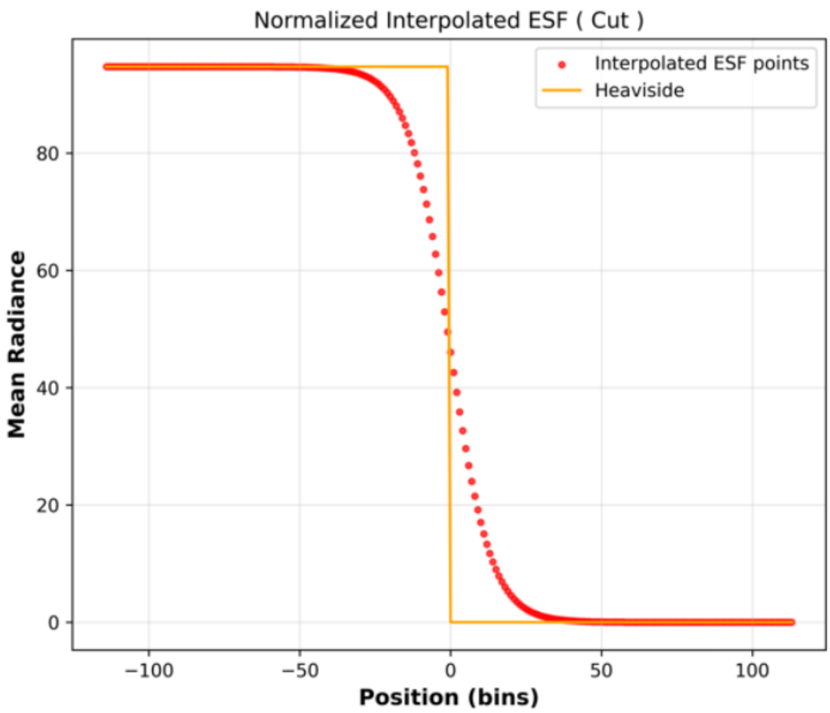


Figure 21 - Interpolated ESF.

This plot displays the interpolated ESF. It is allowed to confirm the accuracy of the interpolation. It is the values used to compute LSF and MTF.

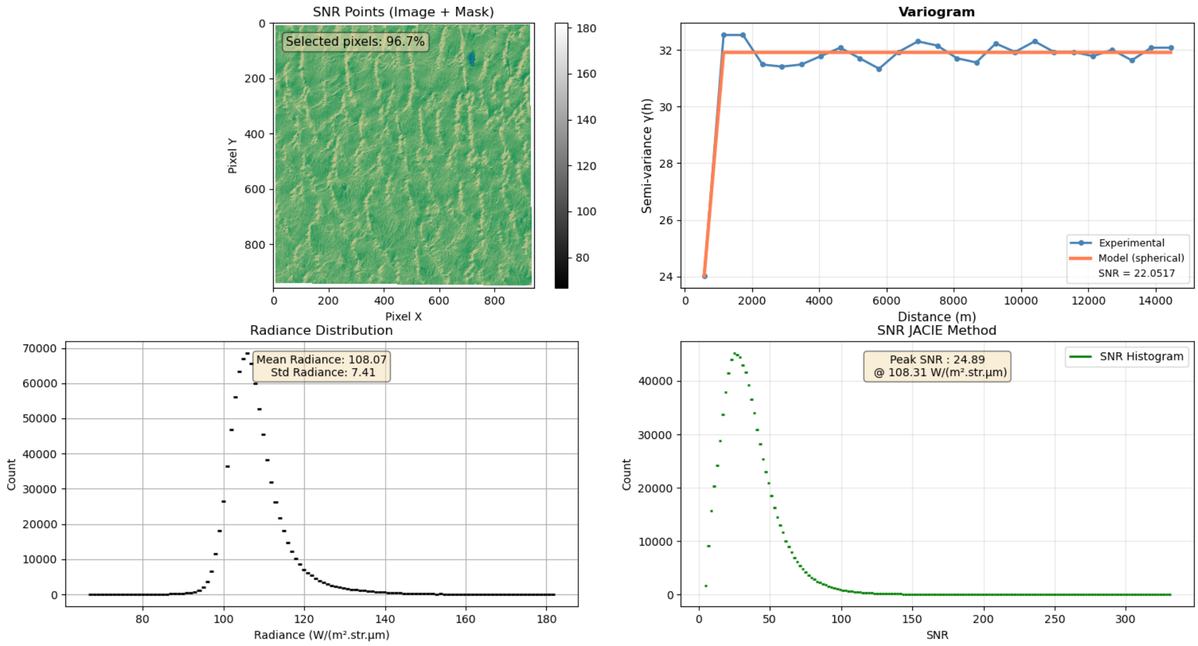
5.1.2.3 GeoPackage

A GeoPackage (.gpkg) is a modern, open-standard geospatial file format designed to store and manage spatial data efficiently in a single, portable file.
The Image Quality Toolset is proposing as output a GeoPackage file. This file it includes following information:

- ROI file (shape file) system Path,
- Input Image file system Path,
- Execution log
- Processing configuration (Input parameters)
- Results

5.2 SNR Processing Report

The SNR Estimator produces a single level of output: a multi-panel figure, displaying the input image with the selected pixels, the variogram fit, the radiance distribution, and the JACIE SNR histogram, allowing immediate inspection of both the intermediate processing steps and the final SNR estimates.



Four graphs are available :

- **Top-left — "SNR Points"**
  Displays the input image in grayscale with the selected uniform pixels overlaid in a terrain colormap (semi-transparent). A text box shows the percentage of pixels retained by the mask. A colorbar shows radiance values.
- **Top-right — "Variogram"**
  Shows the variogram (blue dots/line) and the fitted model. The X-axis is distance in meters, Y-axis is semi-variance γ(h).
- **Bottom-left — "Radiance Distribution"**
  A step histogram of the pixel count per radiance bin (1 W/(m².str.µm) bins) for the selected pixels. A text box shows the mean and standard deviation of the radiance values.
- **Bottom-right — "SNR JACIE Method"**

A step histogram of per-pixel local SNR values, showing the distribution used by the JACIE peak method. A text box annotates the peak SNR value and its corresponding reference radiance.

6. GENERAL COMMAND PARAMETERS

Both the QGIS Processing Toolbox interface and the command-line interface share the same underlying parameters, grouped into three categories: common parameters (used by all methods), MTF-specific parameters, and SNR-specific parameters.

6.1 Common Parameters

These parameters apply to both the MTF Estimator and the SNR Estimator, regardless of the interface used.

Table 4: Common Parameters.

Parameter	Description
Raster layer	Input image layer (any GDAL-supported format: TIFF, etc.)
Band number	Band to process (1-based index). Default: 1
Area of interest	Polygon vector layer (shapefile) defining the ROI

6.1.1 MTF Estimator — Specific Parameters

The following parameters are available only when method = MTF. In the graphical interface, they appear in the MTF Estimator tool panel. In the INI file, they are placed in the [parameters] section alongside the common parameters.

Table 5: Specific Parameters for MTF method.

Parameter	Type	Default	Description
scale	Float	0.01	Conversion scale of the sensor
offset	Float	0.0	Offset correction factor (float). Default: 0.0
px_margin	Integer	1	Number of pixels to exclude at the edges of the ROI during edge detection
edge_direction	Enum	AL	Orientation of the edge: AL (Along Track) or CT (Across Track)
esf_model	Enum	sigmoid	Analytical model for ESF fitting (Parametric only): sigmoid, esf_tanh, esf_fermi, esf_gauss_exp_param

6.1.2 SNR Estimator — Specific Parameters

TO BE COMPLETED

6.2 Interfaces

The toolset is accessible through two interfaces: a graphical interface integrated in QGIS, and a command-line interface driven by INI configuration files. Both interfaces expose the same parameters described in Chapter 6.

6.2.1 Graphical Interface (QGIS Processing Toolbox)

The graphical interface is available directly from the QGIS Processing Toolbox. Parameters map one-to-one to the descriptions in Chapter 6. Enum parameters (Edge direction, ESF interpolation method, ESF model) are presented as drop-down lists. The ESF model drop-down is enabled only when Parametric is selected as the interpolation method.

6.2.2 Command-Line Interface

The CLI uses Python scripts driven by INI configuration files. The IMAGE_QUALITY_TOOL_HOME environment variable must point to the image_quality_toolset directory in your clone.

6.2.3 Running a script

```
python image_quality_toolset/scripts/mtf_computer.py --config_file
<path/to/config.ini>
```

INI file structure

The configuration file is organised in three sections: [input], [parameters], and [debug]. Paths support environment variable expansion (e.g. $IMAGE_QUALITY_TOOL_HOME).

[input] section

Table 6: Input target.

Parameter	Required	Description
image_path	Yes	Absolute path to the input image (TIFF or any GDAL-supported format). Environment variables are expanded.
ROI	No	Region of Interest: path to a shapefile (.shp) or a dict defining a crop window: {'line': <start>, 'pixel': <start>, 'line_number': <h>, 'pixel_number': <w>}. If omitted, the full image is processed.

[parameters] section

Table 7: Input parameters.

Parameter	Required	Description
method	Yes	Algorithm to use: MTF (MTF Estimator) or SNR (SNR Estimator)
band	Yes	Band number to process (1-based)
scale	Yes	Conversion scale (float). Default: 0.01
offset	Yes	Offset correction factor (float). Default: 0.0
px_margin	Yes	Pixel margin for edge extraction (integer). Default: 1
edge_direction	Yes	Edge direction: AL (Along Track) or CT (Across Track)

esf_model	Yes	ESF analytical model: sigmoid, esf_tanh, esf_fermi, esf_gauss_exp_param, esf_erf, loess, esf_to_eq_space_polynomial esf_erf

[debug] section

Table 8: Output directory.

Parameter	Required	Description
enabled	No	Enable debug output (true / false). Default: false
dir	No	Absolute path to the directory where output figures are saved

7. RUNNING THE TOOL FROM PYTHON (COMMAND LINE)

This section describes how to set up and run the Image Quality Toolset from the command line using Python, after cloning the repository.

7.1 PREREQUISITES

- Git installed
- Conda (Miniconda or Anaconda) installed

7.2 CLONE THE REPOSITORY

```
git clone https://gitlab2.telespazio.fr/isl/tim/edap/image-quality-tool
cd image-quality-tool
```

7.3 CREATE AND ACTIVATE THE CONDA ENVIRONMENT

A preconfigured environment file is provided at image_quality_toolset/environment.yml. It installs Python 3.9 and all required dependencies (NumPy, SciPy, Matplotlib, GDAL, QGIS, etc.).

```
conda env create -f image_quality_toolset/environment.yml
conda activate mtf
```

7.4 SET REQUIRED ENVIRONMENT VARIABLES

Two environment variables must be configured so the tool can locate its resources.

IMAGE_QUALITY_TOOL_HOME

This variable must point to the image_quality_toolset directory inside the cloned repository.

```
conda env config vars set IMAGE_QUALITY_TOOL_HOME=/path/to/image-quality-tool/image_quality_toolset
conda deactivate
conda activate mtf
```

Note: Replace /path/to/image-quality-tool with the actual path to your clone. A deactivate/activate cycle is required after setting a conda environment variable.

Verify the variable is set correctly:

```
echo $IMAGE_QUALITY_TOOL_HOME
```

PYTHONPATH

Add IMAGE_QUALITY_TOOL_HOME to your Python path so that internal modules are importable:

```
conda env config vars set PYTHONPATH=${PYTHONPATH}:${IMAGE_QUALITY_TOOL_HOME}
conda deactivate
conda activate mtf
```

7.5 PREPARE A CONFIGURATION FILE

The script is driven by an INI configuration file. The file has three sections: [input], [parameters], and [debug].

Pre-configured examples are available in image_quality_toolset/data/config_files/.

See Chapter 6.2.3 for more information.

7.6 RUN THE SCRIPT

From the root of the cloned repository, with the mtf conda environment active:

```
python image_quality_toolset/scripts/mtf_computer.py --config_file
/path/to/my_config.ini
```

Or using a pre-configured example:

```
python image_quality_toolset/scripts/mtf_computer.py --config_file
image_quality_toolset/data/config_files/pan_target_MTF.ini
```

Results are displayed as Matplotlib figures. If debug is enabled, figures are also saved to the directory specified in [debug] dir.